

Definition and Interpretation of Separable Path-specific Effects With Multiple Ordered Mediators

 Yan-Lin Chen^a and  Sheng-Hsuan Lin^{a,b,c,d}

Abstract: Causal mediation analysis examines the mechanism by which exposure affects outcome via mediators. In contrast to single-mediator scenarios, the presence of multiple ordered mediators introduces complex pathways and corresponding path-specific effects, which are difficult to interpret due to the cross-world counterfactual definition. Path-specific effects also require convoluted and unverifiable assumptions for identification. This article proposes a framework of separable path-specific effects as an extension of the separable effect method to the case of multiple ordered mediators. Compared to the traditional approach, separable path-specific effects can be interpreted as the causal effects of several separated components on the outcome, facilitating a more intuitive understanding of underlying mechanisms. We elucidate the relationship between separable and traditional path-specific effects by demonstrating their equivalence under the individual-level isolation assumptions and identifying both effects under the finest fully randomized causally interpretable structured tree graph (FFRCISTG) model, which inherently makes individual-level isolation assumptions. Moreover, weakening the individual-level isolation assumptions to their population-level counterparts, separable path-specific effects remain identifiable under the FFRCISTG model. Under this causal model, the assumptions for identifying separable path-specific effects can be verified in future experiments, thereby addressing the problem of relying on unverifiable cross-world assumptions in the traditional method. We also discuss how this framework can detect violations of assumptions such as the presence of intermediate confounders and the misspecification of causal order among mediators. In summary, compared with the traditional path-specific effects method, the proposed separable method

provides a more verifiable and interpretable approach for causal multiple mediation analysis.

Keywords: Causal mediation analysis; Causal model; Causally ordered multiple mediators; Path-specific effects; Separable effects; Sequential effect decomposition

(*Epidemiology* 2025;36: 677–685)

Causal mediation analysis aims to understand how an exposure affects an outcome through one or more intermediate variables called mediators.¹ When multiple ordered mediators are involved, the interpretation of pathway decomposition and the required assumptions become more intricate.² In response to this challenge, Daniel et al.² proposed fine-grained natural path-specific effects, which completely decomposes the total effect of exposure on outcome into distinct causal pathways. These effects can be identified under parametric assumptions and subsequently evaluated using sensitivity analyses to justify these assumptions. However, this approach cannot be nonparametrically identified.^{3,4} To address this limitation, two main strategies have been developed. The first is the interventional approach,⁵ which provides an analog to the natural path-specific effects. The second focuses on coarser decompositions, such as the “sequential effect decomposition,”^{2,6,7} in which each path-specific effect captures the influence of the exposure on the outcome through a specific mediator, incorporating all subsequent effects induced by this mediator, whether or not they pass through other downstream mediators. Daniel et al.² first defined the latter in the context of multiple mediators (hereafter referred to as traditional path-specific effects). This study adopts the concept of the second strategy and introduces an alternative definition called separable path-specific effects. The former is beyond the scope of this research, as it focuses on different decomposition strategies, and the sum of those pathways corresponds to the interventional total effect rather than the total effect, and is, therefore, not discussed further. The traditional path-specific effects are nonparametrically identifiable if the data-generating process follows a certain nonparametric structural equation model with independent errors,⁸ which relies on cross-world independence assumptions,⁹ including the necessary conditions of unconfoundedness and the absence of intermediate confounding. However, the

Editor’s Note: A related article appears on p. 686.

Submitted September 24, 2024; accepted May 29, 2025; revised online September 17, 2025

From the ^aInstitute of Statistics, National Yang Ming Chiao Tung University, Hsinchu, Taiwan; ^bInstitute of Data Science and Engineering, National Yang Ming Chiao Tung University, Hsinchu, Taiwan; ^cDepartment of Applied Mathematics, National Dong Hwa University, Hualien, Taiwan; and ^dDepartment of Biochemical and Molecular Medical Sciences, National Dong Hwa University, Hualien, Taiwan.

This study is supported by the grant from National Science and Technology Council of Taiwan (No. MOST 111-2628-M-A49-004-MY3).

Disclosure: The authors report no conflicts of interest.

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

 Supplemental digital content is available through direct URL citations in the HTML and PDF versions of this article (www.epidem.com).

Correspondence: Sheng-Hsuan Lin, Institute of Statistics, National Yang Ming Chiao Tung University, 1001 University Road, Hsinchu 30010, Taiwan. E-mail: shenglin@nycu.edu.tw.

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ISSN: 1044-3983/25/365-677685

DOI: 10.1097/EDE.0000000000001887

traditional path-specific effect method has some drawbacks such as being difficult to interpret operatively because the definition of traditional path-specific effects is based on cross-world counterfactuals, and the cross-world assumptions under nonparametric structural equation model with independent errors are unverifiable.

In causal mediation analysis with a single mediator, natural (in)direct effect (NDE and NIE)¹⁰ has been traditionally treated as the standard definition for (in)direct effect. Recently, the separable direct effect and separable indirect effect^{11,12} have been proposed as an important alternative to the NDE and NIE, offering advantages in verifiability and interpretability. This method is based on the premise that the exposure can potentially be separated into two distinct components in future studies, where one component exerts no population-level effect on the outcome without going through the mediator, and the other has no population-level effect on the mediator. In this framework, the separable indirect effect and separable direct effect are defined as the effects of the former and latter components on the outcome, respectively. Alternative definitions using modified treatments instead of physically separated components have been discussed in the previous literature as well.^{13–15} This approach has two advantages: (a) offering more succinct and intuitive interpretations than the NDE and NIE, and (b) feasible experiments for verifying identification assumptions. However, it has not been extended to the multiple-mediator settings.

Therefore, the objective of this article is to propose a framework to extend separable effect method to the case of multiple mediators, referred to as the separable path-specific effects through this article. These can be simply interpreted as the causal effects of the separated components of exposure on the outcome, if the model assumptions are correctly specified, they share the same interpretation of causal pathway decomposition as the corresponding traditional path-specific effects. In addition, separable path-specific effects possess the advantage of the verification of the identification assumptions in future experiments, which is not feasible for the traditional path-specific effects. In summary, separable path-specific effects address two critical disadvantages of the traditional path-specific effects—interpretability and verifiability.

The remainder of the article is organized as follows. We first review the framework of traditional path-specific effects and its limitations in aspects of interpretability and verifiability. We then propose the framework of separable path-specific

effects and compare both frameworks. Finally, we summarize the findings of this article.

REVIEW OF TRADITIONAL PATH-SPECIFIC EFFECTS

Let A , C , and Y denote the exposure, baseline confounders, and the outcome of interest, respectively; $M_1, \dots, M_K$ are the causally ordered mediators of interest. Causal mediation analysis aims to decompose the causal pathway from exposure to outcome into multiple path-specific effects, which is up to 2^K according to the directed acyclic graph in Figure 1A.¹⁶ For simplicity, we consider the setting of $K = 2$ mediators as shown in Figure 1B, since generalizing to any arbitrary number of mediators is straightforward. In this case, the total effect can be decomposed into four pathways as $A \rightarrow Y$, $A \rightarrow M_1 \rightarrow Y$, $A \rightarrow M_2 \rightarrow Y$, and $A \rightarrow M_1 \rightarrow M_2 \rightarrow Y$, denoted PW_{AY} , PW_{AM_1Y} , PW_{AM_2Y} , and $PW_{AM_1M_2Y}$, respectively. Let $M_1(a)$, $M_2(a)$, and $Y(a)$ denote, respectively, the potential counterfactual values of the mediators and outcome, that is, M_1 , M_2 , and Y , that would be observed had the exposure A been set to a ; $M_2(a, m_1)$ denotes the counterfactual of M_2 would be observed had A been set to a , and M_1 to m_1 ; $Y(a, m_1, m_2)$ denote the counterfactual of Y would be observed had A , M_1 , and M_2 been set to a , m_1 and m_2 , respectively. Connecting these counterfactuals to observed data requires assumptions called consistency and composition. The consistency assumption in this context is that $M_1(a) = M_1$ when $A = a$, $M_2(a, m_1) = M_2$ when $(A, M_1) = (a, m_1)$, and that $Y(a, m_1, m_2) = Y$ when $(A, M_1, M_2) = (a, m_1, m_2)$.¹⁷ The composition assumption is $Y(a, M_1(a), M_2(a, M_1(a))) = Y(a)$.¹⁸ Accordingly, the formal counterfactual definition of these four pathways can be represented as follows:

$$\begin{aligned} TE &\equiv E\{Y(1) - Y(0)\} \\ &= \phi_0(1, 1, 1, 1) - \phi_0(0, 0, 0, 0) \\ &= \{\phi_0(1, 0, 0, 0) - \phi_0(0, 0, 0, 0)\} (PW_{AY}) \\ &\quad + \{\phi_0(1, 1, 1, 0) - \phi_0(1, 0, 1, 0)\} (PW_{AM_1Y}) \\ &\quad + \{\phi_0(1, 0, 1, 0) - \phi_0(1, 0, 0, 0)\} (PW_{AM_2Y}) \\ &\quad + \{\phi_0(1, 1, 1, 1) - \phi_0(1, 1, 1, 0)\} (PW_{AM_1M_2Y}), \end{aligned}$$

Where

$\phi_0(a', a'', a''', a''') \equiv E\{Y(a', M_1(a''), M_2(a''', M_1(a''')))\}$ ^{2,19} is called a mediation parameter. As noted by Daniel et al.,² there are 24 possible decompositions of the total effect in this context, each based on the same mediation parameters. For clarity, we have selected one specific decomposition to

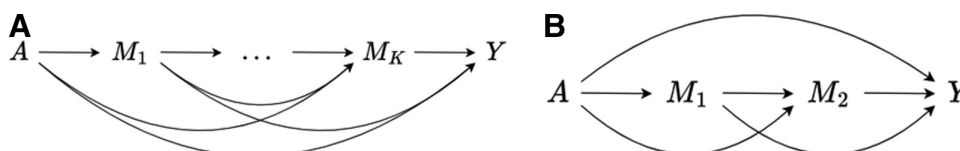


FIGURE 1. Causal directed acyclic graphs (DAGs) for mediation analysis with (A) multiple mediators and (B) two mediators. For simplicity, the measurable baseline confounders C among A , M_1 , ..., M_K , and Y are not shown in the figures.

highlight the pathways of interest. In particular, the effect measures of PW_{AY} and PW_{AM_2Y} are consistent with the path-specific effects proposed by Tai et al.,²⁰ who argued that this effect definition enables the correct classification of mediation–interaction mechanisms; under this definition, PW_{AY} correctly captures the interaction effect that does not pass through the mediators while excluding all mediated interaction,²¹ whereas PW_{AM_2Y} correctly excludes all interaction effects mediated by M_1 . Under this choice, two candidate decompositions remain, and we arbitrarily present one set of effect measures for PW_{AM_1Y} and $PW_{AM_1M_2Y}$, which corresponds to Decomposition 3 in Table 2 of Daniel et al.² This is because, among all 24 decompositions, the effect measures for PW_{AM_1Y} and $PW_{AM_1M_2Y}$ are unidentifiable^{2,3} even under strong models such as nonparametric structural equation model with independent errors (as long as their effect measure definitions involve $\phi_0(a', a'', a''', a''')$ where $a'' \neq a'''$).^{22,23}

Consequently, Steen et al.⁶ extended Daniel et al.’s² work and proposed six possible decompositions of the total effect, each implicitly offering a distinct perspective on the pathways. In this study, we selected one decomposition as a representative example:

Decomposition 1

$$TE = \phi(1, 1, 1) - \phi(0, 0, 0) = PSE_0 + PSE_1 + PSE_2,$$

$$PSE_0 \equiv \phi(1, 0, 0) - \phi(0, 0, 0),$$

$$PSE_1 \equiv \phi(1, 1, 1) - \phi(1, 0, 1),$$

$$PSE_2 \equiv \phi(1, 0, 1) - \phi(1, 0, 0),$$

where $\phi(a', a'', a''') \equiv E\{Y(a', M_1(a''), M_2(a''', M_1(a'')))\}$. In Decomposition 1, PSE_1 capture the pathway that is sum of PW_{AM_1Y} and $PW_{AM_1M_2Y}$ (collectively referred to as $PW_{AM_1 \sim Y}$), and PSE_0 and PSE_2 capture PW_{AY} and PW_{AM_2Y} , respectively. This definition aligns with Tai et al.’s²⁰ proposition discussed in the preceding paragraph and corresponds to the path-specific effects defined by VanderWeele et al.⁷ in the context of mediator–outcome confounders influenced by exposure. That said, other researchers have also utilized the alternative five decompositions, such as those discussed by Huang and Yang.²⁴ This indicates that while we believe the decomposition presented here is optimal, the question of which decomposition is universally optimal remains an open one. Nevertheless, the other five possible decompositions can still be seamlessly applied to our proposed framework. Therefore, in this paper, we focus on this specific decomposition approach while acknowledging the flexibility of the framework to accommodate other decompositions in future research.

The above approach belongs to “sequential effect decomposition,”⁶ which sequentially decomposes the total effect into (a) $PW_{AM_1 \sim Y}$: the impact of the exposure directly influencing the first mediator and then affecting outcome (possibly mediated by M_2), (b) PW_{AM_2Y} : the impact of the exposure directly influencing the second mediator and then affecting outcome,

and (c) PW_{AY} : the direct impact of the exposure on outcome without involving any mediators. Since the pathways to be obtained is less detailed, the mediation parameter $\phi(a', a'', a''')$ can be nonparametrically identified as the mediation formula $Q(a', a'', a''') \equiv E(E[E(Y | M_1, M_2, A = a', C) | M_1, A = a''', C] | A = a'', C)$ under nonparametric structural equation model with independent errors associated with Figure 1B which implies the cross-world independence assumption $A \perp\!\!\!\perp M_1(a') \perp\!\!\!\perp M_2(a'', m_1) \perp\!\!\!\perp Y(a''', m_1, m_2) | C$, for all a', a'', a''', m_1, m_2 .² However, the main disadvantage of the path-specific effect method is that the “cross-world” assumption is based on the independence between variables in distinct counterfactual worlds under intervening exposure with different statuses, making the assumption hardly to be verified. Moreover, each path-specific effect is difficult to interpret in an operational way. As the number of mediators increases, the pathway interpretations of the traditional path-specific effect become even more exaggeratedly complicated. To this end, in the next section, we propose the framework of separable path-specific effects of which the definition does not rely on cross-world counterfactuals, providing a more intuitive interpretation and verifiable assumptions for identification.

SEPARABLE PATH-SPECIFIC EFFECTS

Definition of Separable Path-specific Effects

In the previous section, the effect of exposure on outcome is divided into three pathways. If the path-specific effects are independently caused by some elements of the exposure, then evaluating the impact of these individual elements on the outcome would target the desired path-specific effects. This notion was initially proposed by Robins and Richardson¹¹ and subsequently formalized as the separable effect framework.^{12,13,25} In this section, we extend this work to the context of multiple mediators, referred to as the separable path-specific effects method; the next section discusses the scenarios in which certain assumptions are violated. As a prerequisite for separable path-specific effects, the exposure (A) must be composed of three components (or $K + 1$ components in general): A_Y , A_1 , and A_2 as shown in Figure 2A, which are ideally separable for independent intervention. In the current dataset, an individual is considered (un)exposed to A if and only if this individual is (un)exposed to all three components simultaneously. This deterministic relationship is defined as $A = a' \Leftrightarrow A_Y = A_1 = A_2 = a'$, where $a' \in \{0, 1\}$. When exposure can be hypothetically separated for independent interventions, this allows the definition of the counterfactual variable $W(a_Y = a', a_1 = a'', a_2 = a''')$ as the counterfactual value of $W \in \{M_1, M_2, Y\}$ had (A_Y, A_1, A_2) been intervened upon to take the values (a', a'', a''') , for $a', a'', a''' \in \{0, 1\}$. In this case, there are a total of eight possible joint intervention combinations, where $A = 0$ and $A = 1$ are simply two special cases. In other words, the counterfactual outcome $W(a_Y = a', a_1 = a', a_2 = a')$ is equivalent to $W(a = a')$, for $a' \in \{0, 1\}$. Compared with

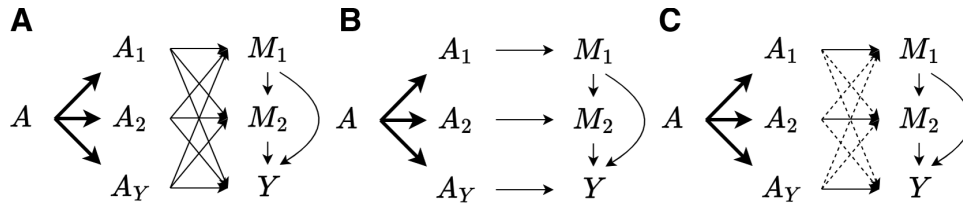


FIGURE 2. A, A causal directed acyclic graph (DAG) for separable path-specific effects. B, A causal DAG based on (A), incorporating the individual-level isolation assumption. C, A causal DAG based on (A), incorporating the population-level isolation assumptions. Dashed edges signify the presence of individual-level effects but the absence of population-level effects. Thick edges indicate that A_1 , A_2 , and A_Y are determined by A . For the sake of simplicity, the measurable baseline confounders C among A , M_1 , M_2 , and Y are not shown in the figure.

previously defined counterfactuals, for example, $Y(a, m_1, m_2)$, which intervenes on A , M_1 , and M_2 , $Y(a_Y, a_1, a_2)$ targets interventions on the three components of exposure A_Y , A_1 , and A_2 . Similarly, we can define the counterfactual $Y(a_Y, a_1, a_2, m_1, m_2)$, where all A_Y , A_1 , A_2 , M_1 , and M_2 are intervened upon. Again, based on deterministic relationship, $Y(a_Y = a', a_1 = a', a_2 = a', m_1, m_2) = Y(a = a', m_1, m_2)$.

We also assume that the counterfactuals under separable effects follow the conventional composition assumption, such that: $Y(a_Y, a_1, a_2) = Y(a_Y, a_1, a_2, m_1 = M_1(a_Y, a_1, a_2), m_2 = M_2(a_Y, a_1, a_2, m_1 = M_1(a_Y, a_1, a_2)))$. Hereafter, we simplify $M_1(a_Y = a', a_1 = a'', a_2 = a''')$, $M_2(a_Y = a', a_1 = a'', a_2 = a''')$, and $Y(a_Y = a', a_1 = a'', a_2 = a''')$ as $M_{1S}(a', a'', a''')$, $M_{2S}(a', a'', a''')$, and $Y_S(a', a'', a''')$, respectively.

Three separable path-specific effects, denoted as SE_0 , SE_1 , and SE_2 , are then defined as follows:

Decomposition 2

$$TE = \phi^{SE}(1, 1, 1) - \phi^{SE}(0, 0, 0) = SE_0 + SE_1 + SE_2,$$

$$SE_0 \equiv \phi^{SE}(1, 0, 0) - \phi^{SE}(0, 0, 0),$$

$$SE_1 \equiv \phi^{SE}(1, 1, 1) - \phi^{SE}(1, 0, 1),$$

$$SE_2 \equiv \phi^{SE}(1, 0, 1) - \phi^{SE}(1, 0, 0),$$

where $\phi^{SE}(a', a'', a''') \equiv E\{Y_S(a', a'', a''')\}$. The separable path-specific effects (SE_0 , SE_1 , and SE_2) can be operatively interpreted as the effects of separated components (A_Y , A_1 , and A_2) on the outcome (Y), respectively. Further, to ensure these separated component effects capture the causal pathways of interest, additional assumptions, termed the dismissible component assumptions,^{12,13,25} are required:

Assumption-DC 1

$$E\{Y_S(a', a'', a''') \mid M_{1S}(a', a'', a'''), M_{2S}(a', a'', a'''), C\} = E\{Y_S(a', a', a') \mid M_{1S}(a', a', a'), M_{2S}(a', a', a'), C\},$$

Assumption-DC 2

$$P\{M_{2S}(a', a'', a''') = m'_2 \mid M_{1S}(a', a'', a'''), C\} =$$

$$P\{M_{2S}(a''', a''', a''') = m'_2 \mid M_{1S}(a''', a''', a'''), C\},$$

Assumption-DC 3

$$P\{M_{1S}(a', a'', a''') = m'_1 \mid C\} = P\{M_{1S}(a'', a'', a''') = m'_1 \mid C\},$$

for all a' , a'' , a''' , m'_1 , and m'_2 . There are four required conditions for holding Assumptions-DC 1–3: (a) the “population-level isolation assumptions,” (b) no unmeasured baseline confounding, (c) no intermediated confounding, and (d) no misspecification of causal order among mediators. In this subsection, we assume conditions (b) to (d) hold, which will be discussed subsequently, and concentrate on the relation between Assumptions-DC 1–3 and the population-level isolation assumptions,^{13,14} that is, the effects represented by the dashed edges in Figure 2C do not exist at the population level. Specifically, Assumption-DC 1 implies no direct effect of A_1 and A_2 on Y ; Assumption-DC 2 implies no direct effect of A_Y and A_1 on M_2 ; Assumption-DC 3 implies no direct effect of A_Y and A_2 on M_1 . Note that the dismissible component assumptions allow the presence of these effects at the individual level, but the population level have to be null. At this point, the average causal effects of A_Y , A_1 , and A_2 on Y capture the desired pathways PW_{AY} , $PW_{AM_1 \sim Y}$, and $PW_{AM_2 Y}$, respectively, coinciding with that of the traditional path-specific effects.

Connection Between the Separable and the Traditional Path-specific Effects

To dive into the relationship between the separable and traditional path-specific effects, let us consider a special case of Assumptions-DC 1–3 called the individual-level isolation assumptions:

Assumption-Iso 1 $Y(a_Y, a_1, a_2, m_1, m_2) = Y(a_Y, m_1, m_2)$,

Assumption-Iso 2 $M_2(a_Y, a_1, a_2, m_1) = M_2(a_2, m_1)$,

Assumption-Iso 3 $M_1(a_Y, a_1, a_2) = M_1(a_1)$,

for all a_Y, a_1, a_2, m_1 , and m_2 . These are stronger assumptions than population-level isolation assumptions, indicating that for each person there are no effects of A_1 and A_2 on Y , no effects of A_Y and A_1 on M_2 , and no effects of A_Y and A_2 on M_1 , as shown in Figure 2B are now absent, then we have the following theorem.

Theorem 1

Decomposition 1 and Decomposition 2 are equivalent under Assumptions-Iso 1–3.

The proof of Theorem 1 is provided in eAppendix A; <https://links.lww.com/EDE/C253>. It is demonstrated that the individual-level isolation assumptions play a crucial role in connecting the separable and the traditional path-specific effects. However, we need to stress that it is precisely in the absence of these assumptions that separable effects demonstrate their advantages by avoiding complex interpretations and unverifiable assumptions. In the preceding subsection, we have shown that Assumptions-DC 1–3 prevent each separable effect from containing effects unrelated to the corresponding pathway, thus appropriately characterizing the mediating roles of these relevant mediators. Nevertheless, further assumptions are required to identify separable path-specific effects in observational studies. In the following theorem, we consider two causal models that are sufficient for this purpose: the finest fully randomized causally interpretable structured tree graph (FFRCISTG) model²⁶ associated with Figure 2B and the FFRCISTG model associated with Figure 2C, which relax the individual-level isolation assumptions to the population-level dismissible component assumptions. Compared with the former, the latter relaxes the individual-level isolation assumptions to population-level isolation assumptions, which imply Assumptions-DC 1–3.

Theorem 2

(a) Under the FFRCISTG model associated with Figure 2B, which implies Assumptions-Iso 1–3 and $A \perp\!\!\!\perp M_1(a) \perp\!\!\!\perp M_2(a, m_1) \perp\!\!\!\perp Y(a, m_1, m_2) | C$, both the mediation parameter $\phi^{SE}(a', a'', a''')$ and $\phi(a', a'', a''')$ are identified as $Q(a', a'', a''')$.

(b) Under the FFRCISTG model associated with Figure 2C, which implies Assumptions-DC 1–3 and $A \perp\!\!\!\perp (M_1(a), M_2(a), Y(a)) | C$, the mediation parameter $\phi^{SE}(a', a'', a''')$ is identified as $Q(a', a'', a''')$.

The proof of Theorem 2 is provided in eAppendix B; <https://links.lww.com/EDE/C253>. At this point, we have introduced the definitions and identification conditions for both traditional and separable path-specific effects, summarized in the upper half of eTable 1 in eAppendix G; <https://links.lww.com/EDE/C253> for ease of comparison. In the next subsection, we demonstrate that the assumptions in the FFRCISTG model associated with Figure 2C are verifiable.

Before proceeding, Table 1 contrasts individual-level isolation assumptions with dismissible component assumptions. Although the individual-level isolation assumptions allow separable path-specific effects to represent causal pathways of interest and equate them with traditional effects, they have notable drawbacks. First, they are unverifiable, losing the key advantage of the separable path-specific effect method, which allows empirical testing and future verification through RCTs. Additionally, its assumptions can also be empirically tested. However, when using the individual-level isolation assumption, these critical advantages of verification are entirely lost. Second, individual-level isolation assumptions are redundant for identification of separable path-specific effects. Specifically, FFRCISTG

models with individual-level isolation assumptions imply both Assumptions-DC 1–3 along with the assumption of $A \perp\!\!\!\perp (M_1(a), M_2(a), Y(a)) | C$ and cross-world independence $A \perp\!\!\!\perp M_1(a') \perp\!\!\!\perp M_2(a'', m_1) \perp\!\!\!\perp Y(a''', m_1, m_2) | C$. These assumptions enable the identification of separable and traditional path-specific effects, respectively. However, as shown in Theorem 2, it is sufficient to identify separable effects by simply using the FFRCISTG models with population-level isolation assumptions. In contrast, dismissible component assumptions offer several advantages: they are verifiable, imply population-level isolation assumptions (providing separable path-specific effects with an intuitive path interpretation), and allow identifiability of separable effects when $A \perp\!\!\!\perp (M_1(a), M_2(a), Y(a)) | C$ is plausible, separable effects become identifiable. In Theorem 1, we emphasize the individual-level isolation assumptions primarily to help readers intuitively understand the relationship between separable and traditional path-specific effects, as well as how population-level isolation relaxes such strict constraints. For the remainder of this article, however, we focus on the dismissible component assumption and their implications.

Verification of Causal Assumptions

To begin with, if the components of the exposure can be intervened upon separately, then Assumptions-DC 1–3 can be verified in an eight-arm randomized controlled trial by assessing: (1) the independence between Y and (A_1, A_2) conditioning on A_Y, M_1, M_2 , and C , (2) the independence between M_2 and (A_Y, A_1) conditioning on A_2, M_1 , and C , and (3) the independence between M_1 and (A_Y, A_2) conditioning on A_1 and C . Once these verifications are passed, it indicates that (a) the population-level isolation assumptions, (b) no unmeasured baseline confounding, (c) no intermediated confounding, and (d) no misspecification of causal order among mediators are ensured to hold. Table 2 outlines which independence tests among (1)–(3) can detect violations of the conditions illustrated in Figure 3A–D. In practice, ensuring that conditions (a)–(d) are fully satisfied is often challenging. Table 2 serves as a guide to identify which condition(s) might be violated based on the results of future independence tests. For instance, if independence tests (2) and (3) fail, this may indicate a misspecification of the causal order among mediators. Alternatively, if only test (2) fails, we can rule out the possibility of a misspecification of causal order among mediators and instead consider that A_Y or A_1 might have direct effects on M_2 , or that there might be unmeasured M_1 – M_2 confounders. eTable 2 in eAppendix G; <https://links.lww.com/EDE/C253>, which extends Table 2, provides a detailed listing of the pathways that result in failed independence tests under scenarios where conditions (a)–(d) are violated.

In particular, as the presence of intermediate confounders,^{3,7,9,10,27} which are mediator–outcome confounders or mediator-mediator confounders affected by exposure,

TABLE 1. Comparison of Dismissible Component Assumptions and Individual-level Isolation Assumptions, Including Their Definitions, Verifiability, and Properties Related to Separable and Traditional Path-specific Effects

Dismissible Component Assumptions		Individual-level Isolation Assumptions
Definitions	Assumption-DC 1 $E\{Y_S(d', a'', a'''), M_{2S}(d', a'', a'''), C\} = E\{Y_S(d', a', d'), M_{1S}(d', d', a'), M_{2S}(a', d', a'), C\}$ Assumption-DC 2 $P\{M_{2S}(d', a'', a''') = m'_2 \mid M_{1S}(d', a'', a'''), C\} = P\{M_{2S}(a'', a''', a'''), C\}$ Assumption-DC 3 $P\{M_{1S}(d', a'', a''') = m'_1 \mid C\} = P\{M_{1S}(d', a'', a'') = m'_1 \mid C\}$	Assumption-Iso 1 $Y(a_Y, a_1, a_2, m_1, m_2) = Y(a_Y, m_1, m_2)$ Assumption-Iso 2 $M_2(a_Y, a_1, a_2, m_1) = M_2(a_2, m_1)$ Assumption-Iso 3 $M_1(a_Y, a_1, a_2) = a_1$
Verifiability	Yes	No
Properties	- Implies population-level isolation assumptions, enabling intuitive pathway interpretation for separable PSEs (though separable PSEs do not necessarily equal traditional PSEs). - By assuming $A \perp (M_1(a), M_2(a), Y(a)) \mid C$, separable PSEs can be identified.	- Aligns separable PSEs with traditional PSEs but does not allow identification of separable PSEs or traditional PSEs in terms of observed data alone. - When combined with the FFRCISTG model: (1) Separable PSEs = Traditional PSEs and are identifiable (as shown in Table 2); (2) Implies Assumptions-DC 1–3 and $A \perp (M_1(a), M_2(a), Y(a)) \mid C$; (3) imply cross-world independence assumptions: $A \perp M_1(a') \perp M_2(a'', m_1) \perp Y(a'', m_1, m_2) \mid C$.
FFRCISTG indicates finest fully randomized causally interpretable structured tree graph; PSE, path-specific effects.		

TABLE 2. The Relationship Between the Necessary Conditions (Labeled (a)–(d) on the Left) to Satisfy Assumptions-DC 1–3 and the Corresponding Independence Conditions (Labeled (1)–(3) at the Top) Used to Verify Assumptions-DC 1–3 in Future Experiments

	(0) $Y \perp\!\!\!\perp (A_1, A_2) A_Y, M_1, M_2, C$	(2) $M_2 \perp\!\!\!\perp (A_Y, A_1) A_2, M_1, C$	(3) $M_1 \perp\!\!\!\perp (A_Y, A_2) A_1, C$
(a) Population-level isolation assumptions			
(a-1) No population-level direct effect of A_1 on Y	○		
(a-2) No population-level direct effect of A_2 on Y	○		
(a-3) No population-level direct effect of A_Y on M_2		○	
(a-4) No population-level direct effect of A_1 on M_2		○	
(a-5) No population-level direct effect of A_Y on M_1			○
(a-6) No population-level direct effect of A_2 on M_1			○
(b) No unmeasured baseline confounding			
(b-1) No unmeasured M_1 – M_2 confounder		○	
(b-2) No unmeasured M_1 – Y confounder	○		
(b-3) No unmeasured M_2 – Y confounder	○		
(c) No intermediated confounding			
(c-1) No intermediate confounder L_1	○	○	○
(c-2) No intermediate confounder L_2	○	○	
(c-3) No both intermediate confounders L_1 and L_2	○	○	○
(d) No misspecification of causal order among mediators		○	○

A circle (○) indicates that, under a scenario where the necessary condition on the left is violated (as shown in Figure 3A–D), the independence assumption above the circle will also be violated.

such as L_1 and L_2 in Figure 3C and the reversal of the causal order of mediators, as illustrated in Figure 3D, are relatively common in multiple mediation analyses and are more complex. The strategy for dressing these scenarios is elaborated in eAppendix C; <https://links.lww.com/EDE/C253>. When intermediate confounders are present, the causal structure must meet stricter requirements to identify separable path-specific effects. Specifically, each intermediate confounder can only be directly influenced by a single separated component of the exposure. eFigure 4; <https://links.lww.com/EDE/C253> illustrates four potential causal structures in the presence of intermediate confounders, under which the separable path-specific effects are identifiable. In eAppendix G; <https://links.lww.com/EDE/C253>, eFigure 4A; <https://links.lww.com/EDE/C253> depicts the case where the intermediate confounders are influenced by A_1 , and the sequential effect through L_1 and L_2 is categorized as SE_1 . eFigure 4B; <https://links.lww.com/EDE/C253> shows the scenario where the confounders are affected by A_2 , and the corresponding sequential effect is categorized as SE_2 . eFigure 4C; <https://links.lww.com/EDE/C253> presents the setting in which A_Y influences the confounders, with the sequential effect categorized as SE_0 . In eFigure 4A; <https://links.lww.com/EDE/C253> separable path-specific effects treat L_1 , L_2 , and M_1 as joint mediators. Conversely, eFigure 4D; <https://links.lww.com/EDE/C253> considers L_1 and L_2 as joint mediators but isolates the sequential effects of the exposure through L_1 and L_2 as an independent path-specific effect. The assumptions for identifying these four structures, along with the

corresponding identification results, are detailed in eAppendix D; <https://links.lww.com/EDE/C253>. Notably, the selection among these four methods should be based on the assumptions that are applicable to the specific example under analysis.

Furthermore, the independence $A \perp\!\!\!\perp (M_1(a), M_2(a), Y(a)) | C$ can be guaranteed to hold if the present data are sampled from a randomized trial in which there are no exposure-mediator confounders or exposure-outcome confounders. Alternatively, the independence can still be verified if the natural value of the exposure before the intervention is observable, which need not be an eight-arm trial.

Causal Interpretation of Separable Path-specific Effects

To illustrate the causal interpretation of separable path-specific effects, we use coronavirus disease 2019 (COVID-19) as an example. During the COVID-19 pandemic, the impact of confirmation of COVID-19 infection status on the risk of death includes indirect effects mediated by the “post-infection isolation” and “depression.” Let A represent the confirmation of COVID-19 infection status, M_1 be isolation precaution, M_2 be depression status, and Y stand for death. The total effect of confirmation of COVID-19 infection status (A) on the risk of death is decomposed into SE_1 , SE_2 , and SE_0 , that is, the effects of the influence caused by a positive viral test result (A_1), the impact of the virus on the nervous system (A_2), and the impact of the virus on the respiratory system and other physiologic systems (A_Y), respectively. As shown in

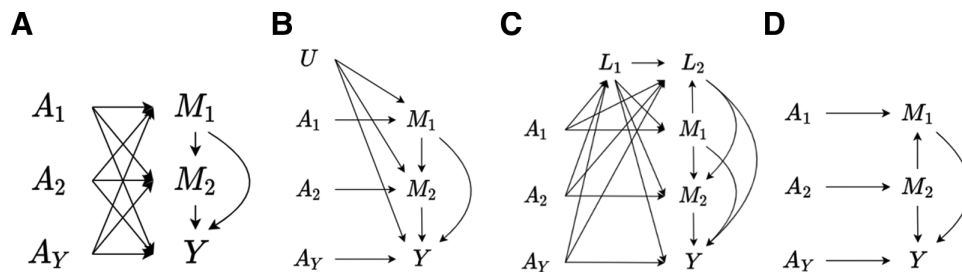


FIGURE 3. Causal directed acyclic graphs (DAGs) in future 8-arm randomized trials. A, A causal DAG that violates the population-level isolation assumptions. B, A causal DAG in the presence of unmeasured baseline confounders U ; C, A causal DAG in the presence of intermediate confounders L_1 and L_2 affected by A . D, A causal DAG with unexpected causal ordering of M_1 and M_2 .

eFigure 5; <https://links.lww.com/EDE/C253>, if (a) the effect of a positive viral test result on death must go through isolation precautions, if (b) the impact of the virus on the nervous system affecting death must go through depression, and if (c) the impact of the virus on the respiratory system and other physiologic systems affects death without passing through isolation precautions or depression, then SE_1 , SE_2 , and SE_0 capture the pathways of interest: $PW_{AM_1 \sim Y}$, $PW_{AM_2 Y}$, and PW_{AY} . It is important to emphasize that the above discussion refers to population-level isolation assumptions. While certain pathways may exist at the individual level, they are considered absent at the population level due to aggregate cancelation. For instance, following a COVID-19 diagnosis, some individuals might experience dizziness or restlessness caused by the virus's impact on the nervous system (A_2), which could lead them to seek relief by stepping outside to access improved air quality or ventilation. In contrast, others might prefer to stay home and adopt self-initiated isolation precautions. However, on average, it is generally recognized that A_2 does not have a population-level effect on whether an individual implements isolation precautions.

Even though both separable and traditional path-specific effects are sequential effect decomposition, there is a discrepancy between their interpretations of the pathways. The latter half of eTable 1; <https://links.lww.com/EDE/C253> compares their interpretations in the context of COVID-19 example. The result presented in eTable 1; <https://links.lww.com/EDE/C253> is based on the following models:

$$\text{logit}(P(M_1 = 1 | A = a)) = -0.847 + 3.044a,$$

$$\begin{aligned} \text{logit}(P(M_2 = 1 | M_1 = m_1, A = a)) \\ = -3.178 + 0.738m_1 + 2.035a, \end{aligned}$$

$$\begin{aligned} \text{logit}(P(Y = 1 | M_1 = m_1, M_2 = m_2, A = a)) \\ = -7.375 - 0.449m_1 + 0.694m_2 + 2.525a, \end{aligned}$$

with parameters chosen to reflect real-world conditions.^{28–36} Detailed descriptions are provided in eAppendix F; <https://links.lww.com/EDE/C253>. The analysis of separable path-specific effects based on our specified data-generating models clearly reveals that the respiratory and other physiologic

impacts (A_Y) of the virus have a much greater impact on survival outcomes compared with the other two separated components. However, for infected individuals, a positive COVID-19 test result (A_1) can substantially reduce mortality rates, likely due to receiving proper care following isolation. Meanwhile, the neurologic damages (A_2) caused by COVID-19 contribute to a slight increase in mortality risk. Unlike traditional path-specific effects in which the use of cross-world counterfactuals makes their interpretation overly abstract, separable effects are constructed under single-world counterfactuals, and thus, it is easier to conceive of causal pathways captured by the effects of the separated components.

DISCUSSION

This article presents an extension of the separable effect framework to encompass the context of causally ordered multiple mediators. We have demonstrated how this framework can be utilized to identify causal pathways from observed data. One advantage of this approach is its straightforward interpretations. This substantially improves upon the limitations of the traditional path-specific effect approach, which employs cross-world counterfactuals. Additionally, we elucidated the relationship between separable and traditional path-specific effects. It was demonstrated that, under rigorous constraints on the causal structure (i.e., when individual-level isolation assumptions are satisfied), the separable and traditional effects are mathematically equivalent. Remarkably, separable path-specific effects offer a more intuitive interpretation, and such interpretations remain valid even in the absence of individual-level isolation assumptions.

Another advantage of our approach is that the identification assumptions can be verified in future experiments, which addresses a significant limitation of the traditional path-specific effects. Table 2 presents four scenarios in which there are violations of the necessary conditions of Assumptions-DC 1–3: (a) the population-level isolation assumptions, (b) no unmeasured baseline confounding, (c) no intermediated confounding, and (d) no misspecification of causal order among mediators, and shows how such violations can be detected. We also discussed the scenarios where intermediate confounders are present and where the causal order of mediators

is uncertain, which are critical challenges in mediation analysis, particularly with multiple mediators. In such cases, the traditional path-specific effect method often falls short due to unverifiable causal structures and assumptions.

Nevertheless, the separable path-specific effect method requires the assumption that the exposure can be separated into several components, which necessitates an in-depth understanding of the nature of the exposure and thereby restricts the scope of causal mediation analysis. Although Stensrud et al.^{13,14} proposed that the modified treatment (exposure) assumption could relax such restriction, allowing separable effects to be defined even when the exposure cannot be physically separated, this assumption is at the individual level and therefore unverifiable.

In summary, as the number of mediators increases, concerns about the validity of assumptions for causal effects also grow. Therefore, methods with verifiable assumptions, such as the separable effect method, are of great importance. In this article, we propose the separable path-specific effects, conduct a detailed investigation of their properties, and compare the limitations encountered by the traditional path-specific effect method in the presence of multiple mediators. We demonstrate that separable effects offer better interpretability and verifiability than traditional path-specific effects in causal mediation analysis with multiple mediators.

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